

Cybersecurity Incident Timeline

Security Awareness & Incident Analysis Report

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Bootcamp: Upskills Cybersecurity

Course Assignment: Lab 1 – Cybersecurity Incident Timeline

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Executive Summary

Cybersecurity incidents have continuously shaped the evolution of modern defensive strategies, security frameworks, compliance standards, and incident response methodologies. This report analyzes five historically significant cyberattacks between 1988 and 2020, identifying their attack methods, business impact, failed security controls, and the security improvements that emerged afterward.

The attacks included in this report are:

Morris Worm (1988)

ILOVEYOU Worm (2000)

Target Breach (2013)

WannaCry Ransomware (2017)

SolarWinds Supply Chain Attack (2020)

The objective of this project is to understand how cybersecurity failures occur, how attackers exploit weak controls, and how organizations adapt by implementing stronger security practices.

Introduction

Cybersecurity incidents are rarely isolated technical failures. Most successful attacks occur because of weak processes, poor visibility, lack of patching, inadequate monitoring, insecure configurations, or insufficient employee awareness.

By studying major cyberattacks across multiple decades, security professionals can identify recurring patterns and improve defensive strategies. This report presents a timeline-based analysis of five major attacks and explains how each incident influenced modern cybersecurity controls and governance.

Research Table – Major Cybersecurity Attacks

Visual Timeline

Analysis Questions

Q1: Which two attacks had the most lasting impact on modern security practice? Why?

The two attacks with the most lasting impact on modern cybersecurity practices were the WannaCry ransomware attack and the SolarWinds supply chain attack. Security patches and zero trust models are taken basic and most important security measures as they can escalate in global infrastructure crisis if not taken seriously in present days.

Q2: What common theme do you see across at least three attacks? Which security principle was violated?

A common theme across the Morris Worm, ILOVEYOU, and WannaCry attacks was the rapid self-propagation of malware across connected systems. These attacks spread widely because organizations lacked proper patch management, network segmentation, and security awareness. The security principles violated included defense in depth and availability.

Q3: Target Breach – Which NIST Cybersecurity Framework (CSF) function was most lacking?

The most lacking NIST Cybersecurity Framework function in the Target Breach was Protect. Although the attackers initially entered through stolen third-party vendor credentials, the breach became catastrophic because Target lacked sufficient network segmentation and access controls to protect sensitive payment systems.

Q4: Add One More Attack Not on the List

Equifax Breach (2017)

Attack Type: Web Application Breach / Data Theft

Impact: Personal data of approximately 147 million individuals was exposed, including Social Security numbers and financial information.

What Failed: Failure to patch a known Apache Struts vulnerability, weak vulnerability management, and inadequate asset visibility.

Lesson Learned: Organizations must maintain continuous patch management, accurate asset inventories, and proactive vulnerability monitoring.

Key Lessons Learned

The attacks analyzed in this report reveal several recurring cybersecurity lessons:

Unpatched systems remain one of the most dangerous organizational weaknesses.

Human error and social engineering continue to be highly effective attack vectors.

Third-party vendors can introduce major security risks.

Network segmentation and least privilege are critical for limiting attacker movement.

Continuous monitoring and rapid incident response are essential for minimizing damage.

Modern organizations must adopt proactive security models rather than relying solely on perimeter defenses.

Conclusion

Cybersecurity has evolved significantly over the past several decades, but the core causes of major breaches remain surprisingly consistent. Weak security practices, inadequate monitoring, poor patch management, and excessive trust continue to enable attackers.

Each attack discussed in this report influenced the development of modern cybersecurity controls, frameworks, and operational strategies. By understanding these historical incidents, organizations can better prepare for future threats and strengthen their overall security posture.

The evolution from the Morris Worm to SolarWinds demonstrates that cybersecurity is no longer only a technical issue — it is a business, operational, and national security priority.

References

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